An Automatic Following/Forming Framework for Inverter-Based Resources

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Table of Contents

Nomenclature

Abbreviations

EMF	electromotive force
GFL / GFM	grid-following / grid-forming control
IBR	inverter-based resource
PF / TS	power flow / time-domain simulation
PLL	phase-locked loop
SG	synchronous generator
VSG	virtual synchronous generator

Symbols

D_v, M	VSG damping / virtual inertia constant
E	GFM internal voltage magnitude reference
E'_q, E'_d	SG transient internal EMFs (q/d-axis)
E''_q, E''_d	SG subtransient internal EMFs (q/d-axis)
F_0, F_1	IBR vector fields, GFL / GFM branch
f_0, f_{coi}, ω_b	nominal / centre-of-inertia frequency; $\omega_b = 2\pi f_0$
$g = 0$	network current balance at every bus, PF and TS
I_{dc}, V_{dc}	converter DC-link current / voltage
i_d, i_q	converter filter currents (d/q-axis)
k_p, k_i	PI proportional / integral gains
m_i	IBR i mode: 0 = GFL, 1 = GFM
P_j, Q_j	net active / reactive injection at bus j
t	time
u	exogenous inputs: set-points, loads, events
V_i, θ_i	bus i voltage magnitude / phase angle
x	full differential state vector
x_{SG}, x_{IBR}	SG / IBR state vectors
Y	network admittance matrix
y	algebraic variables: bus voltages
δ, ω	SG rotor angle / speed
θ_{PLL}, ξ_{PLL}	PLL estimated angle / integrator state
θ_{VSG}, ω_{VSG}	VSG internal angle / angular speed
ξ_P, ξ_Q	power-loop integrator states
ξ^b_{Id}, ξ^b_{Iq}	current-loop integrators, $b \in \{GFL, GFM\}$
ξ_{Vd}, ξ_{Vq}	voltage-loop integrator states

Introduction

- Increasing IBR penetration.
- Less synchronous-machine inertia.
- Weak grids challenge PLL-based GFL control.
- GFM control provides voltage and frequency support.
- Adapt the control mode to grid conditions [1], [2].

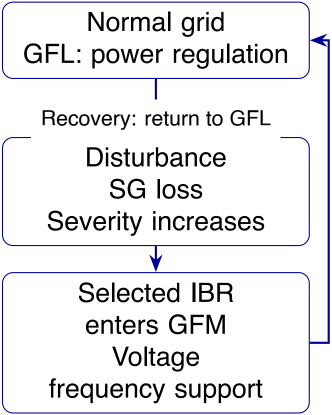

Figure 1: GFL/GFM mode-switching concept.

System Description and Operating Principles

PF
Rotor and converter states are constant; bus currents balance.

$$0 = g(x, y, u_0, m). \tag{1}$$

$$P_i^{inj} - P_j(x, y, u_0, m) = 0, \tag{2}$$

$$Q_i^{inj} - Q_j(x, y, u_0, m) = 0. \tag{3}$$

TS
Machine and converter states respond to disturbances.

$$\dot{x} = f(x, y, u, m, t), \tag{4}$$

$$0 = g(x, y, u, m, t), \tag{5}$$

$$m_i = \begin{cases} 0 & \text{GFL}, \\ 1 & \text{GFM}. \end{cases} \tag{6}$$

$$x = [x_{SG}^T, x_{IBR,1}^T, \ldots, x_{IBR,4}^T]^T, \tag{7}$$

$$x_{SG} = [\delta, \omega, E'_q, E'_d, E''_q, E''_d]^T, \tag{8}$$

$$x_{IBR} = [x_p^T, x_{GFL}^T, x_{GFM}^T] \tag{9}$$

Shared physical states

$$x_p = [i_d, i_q, V_{dc}, I_{dc}]^T \qquad (10)$$

$$\frac{L_f}{\omega_b}\dot{i}_d = v_{cd} - v_d - R_f i_d + \omega_{pu} L_f i_q, \qquad (11)$$

$$\frac{L_f}{\omega_b}\dot{i}_q = v_{cq} - v_q - R_f i_q - \omega_{pu} L_f i_d. \qquad (12)$$

$$\omega_{pu} = \begin{cases} 1 + \dot{\theta}_{PLL}/\omega_b & m_i = 0, \\ \omega_{VSG} & m_i = 1. \end{cases} \qquad (13)$$

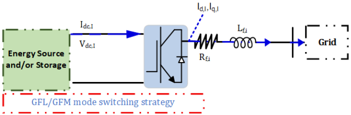

Figure 2: IBR model.

Switchable GFL/GFM Control Model

Controller state vectors

$$x_{GFL} = [\theta_{PLL}, \xi_{PLL}, \xi_P, \xi_Q, \xi_{Id}^{GFL}, \xi_{Iq}^{GFL}]^T \tag{15}$$

$$x_{GFM} = [\theta_{VSG}, \omega_{VSG}, E, \xi_{Vd}, \xi_{Vq}, \xi_{Id}^{GFM}, \xi_{Iq}^{GFM}]^T \tag{16}$$

GFL: PLL + power control

$$\dot{\theta}_{PLL} = k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL} \tag{17}$$

GFM: VSG + voltage control

$$M\dot{\omega}_{VSG} = \kappa P_{ref} - P_{inv} - D_v(\omega_{VSG} - 1) \tag{18}$$

$$\kappa = S_{base}/M_{base} \tag{19}$$

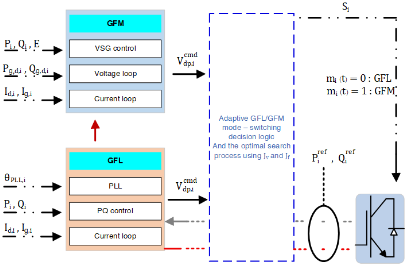

Figure 3: Switchable controller branches.

Mode Switching: Supervisory Readiness

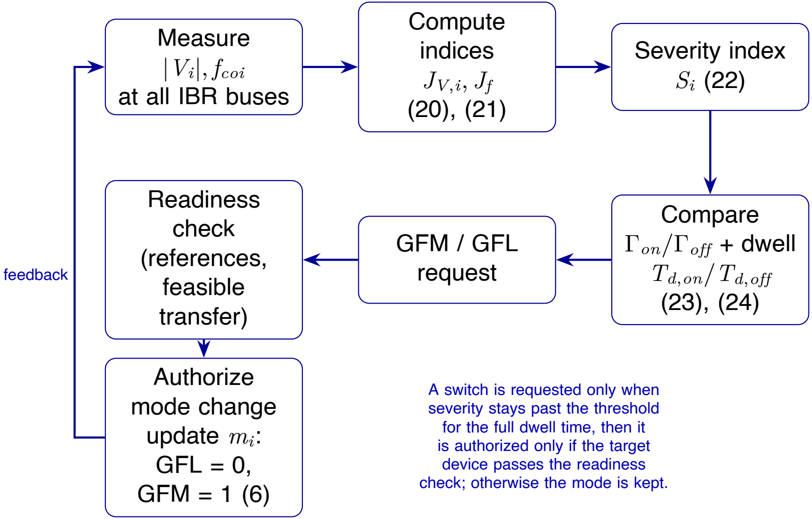

Figure 4: Supervisory readiness and GFL/GFM mode-switching procedure.

Mode Switching: Index and Logic

Voltage and frequency indices

$$J_{V,i} = \frac{\big| |V_i| - V_i^\star \big|}{\Delta V_{base}}, \tag{20}$$

$$J_f = \frac{|f_{coi} - f_0|}{\Delta f_{base}}, \tag{21}$$

$$S_i = \mathrm{sat}_{[0,1]}(w_V J_{V,i} + w_f J_f). \tag{22}$$

$J_{V,i}$, J_f: voltage / centre-of-inertia indices; $S_i \in [0,1]$: severity; $V_i^\star$: healthy per-bus PF magnitude; $\Gamma_{on}, \Gamma_{off}$: thresholds; $T_{d,on}$, $T_{d,off}$: dwells; w_V, w_f: weights; $\mathcal{L}, \mathcal{G}$: online GFL/GFM sets.

$$\text{GFM request:} \quad \exists i \in \mathcal{L} : \; S_i \geq \Gamma_{on} \quad \text{for } T_{d,on} \tag{23}$$

$$\text{GFL request:} \quad \forall i \in \mathcal{L} \cup \mathcal{G} : \; S_i < \Gamma_{off} \quad \text{for } T_{d,off} \tag{24}$$

Γ_{on}/Γ_{off}: 0.65/0.35; dwell: 0.10/1.00 s.

Index bases: 0.10 pu, 0.50 Hz; voltage/frequency weights: 0.5/0.5.

Case Study: IEEE 14-Bus System

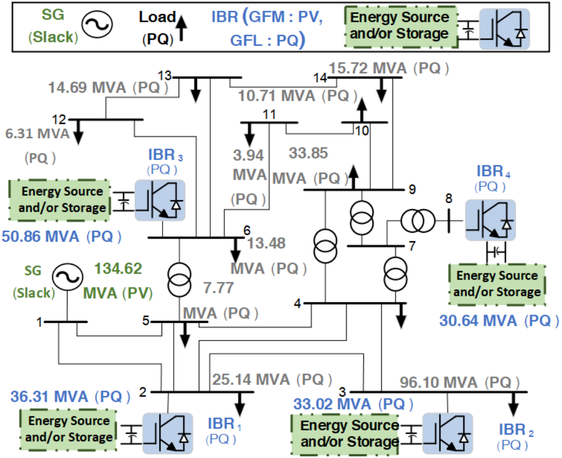

Figure 5: IEEE 14-bus study network [3].

SG1: Bus 1

IBR_1: Bus 2
IBR_2: Bus 3
IBR_3: Bus 6
IBR_4: Bus 8

100 MVA
60 Hz

Parameter Settings: Network and Plant

Table 1: Network and plant parameters.

Parameter	Symbol	Value	Unit
Network and synchronous machine			
System bases	S_{base}, f_0	100,60	MVA, Hz
SG (system base)	H, D	2.50,1.00	s, pu
SG time constants	$T'_{d0}, T''_{d0}, T''_{q0}$	7.40,0.03,0.033	s
IBR plant			
Filter	L_f, R_f	0.15,0.015	pu
DC bus / current limit	$C_{dc}, V^0_{dc}, I_{max}$	0.10,1.00,1.20	pu
DC source	$\epsilon_{dc}, P_r, \tau_s, \zeta^\star$	0.10,1.00,0.005,0.707	–, pu, s, –
Chopper	$V^{max}_{dc}, \delta_{ch}$	1.10,0.02	pu, –

* Table References: [2], [3]

Parameter Settings: Control and Supervisor

Table 2: Controller and supervisor parameters.

Parameter	Symbol	Value	Unit
GFL control			
PLL PI gains	$k_{p,PLL}, k_{i,PLL}$	1.20, 5.00	*
Power PI gains	$k_{p,P/Q}, k_{i,P/Q}$	0.80, 2.50	*
Shared current loop			
Current PI	$k_{p,I}, k_{i,I}$	0.30,4.00	*
GFM control			
VSG	M, D_v	0.08, 20.0	pu†
Q–V loop	k_Q, k_E, τ_E	0.25,8.00,0.05	*, *, s
Voltage PI	$k_{p,V}, k_{i,V}$	1.20, 4.50	*
Supervisor			
Hysteresis	$\Gamma_{on}, \Gamma_{off}$	0.65,0.35	–
Dwell / weights	$T_{d,on/off}; w_V, w_f$	0.10/1.00;0.5,.50	s; –
Hold / lock	T_{hold}, T_{lock}	1.000, 2.000	s
Index bases	$\Delta V_{base}, \Delta f_{base}$	0.10,0.50	pu, Hz

*Table References: [2]

† Report notation: $M = 2H_v$, $H_v = 0.04$ s.

Scenario Timeline

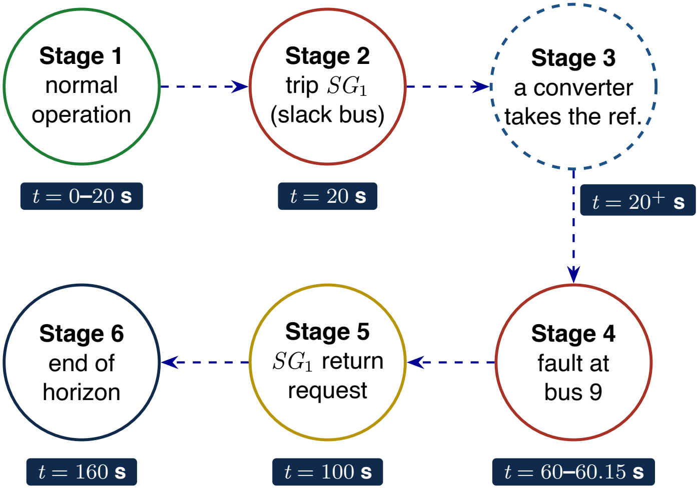

Figure 6: Scenario timeline.

Result: Severity Index and Converter Mode

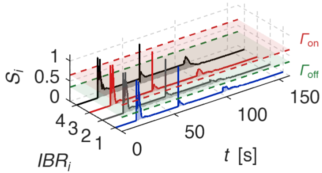

Figure 7: Severity index S_i .

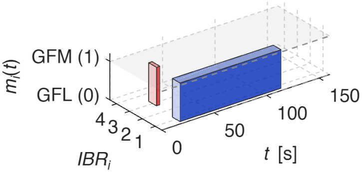

Figure 8: m_i GFL = 0, GFM = 1.

Result: IBR State Response

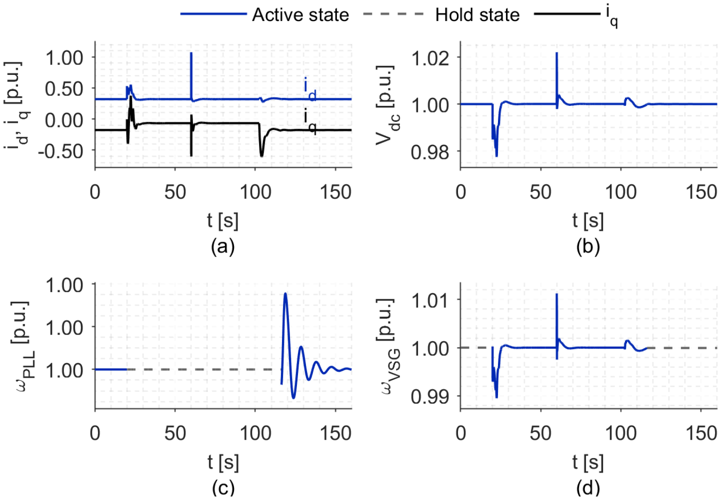

Figure 9: $(a)\ i_d, i_q$; $(b)\ V_{dc}$; $(c)\ \omega_{PLL}$; $(d)\ \omega_{VSG}$

Result: Voltage and Frequency

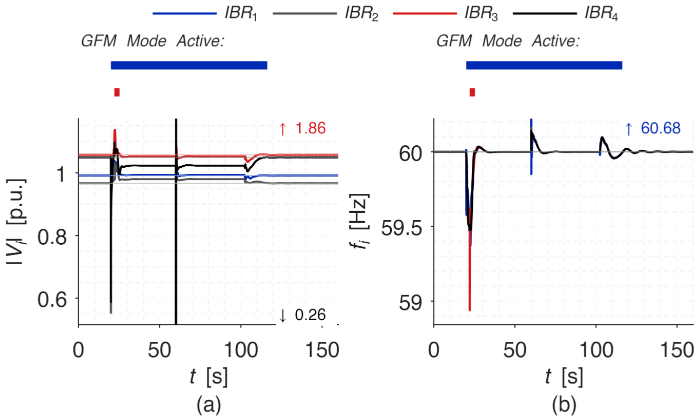

Figure 10: (a) bus-voltage magnitude and (b) converter frequency.

Result: Active and Reactive Power

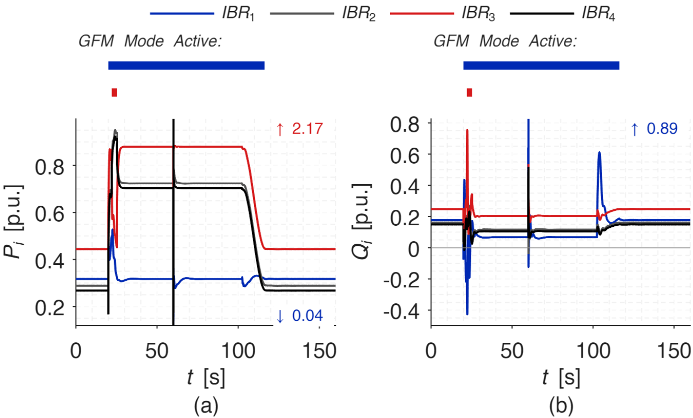

Figure 11: (a) active power and (b) reactive power.

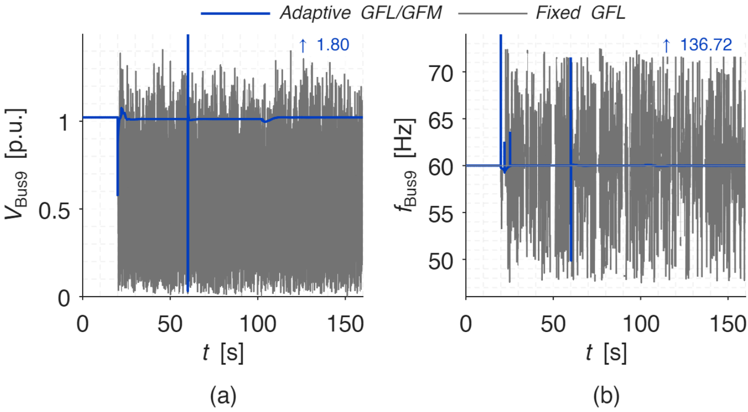

Figure 12: (a) voltage magnitude and (b) frequency.

Fault Bus Active and Reactive Power

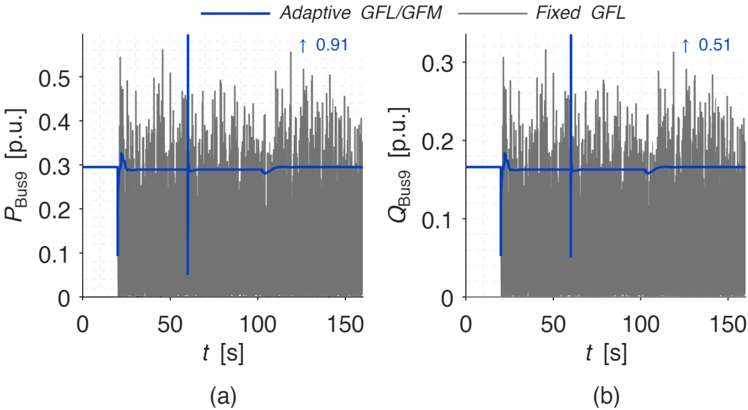

Figure 13: (a) active power and (b) reactive power.

Summary

Completed Development

- ☑ Power flow and time-domain simulation.
- ☑ Switchable GFL/GFM converter model.
- ☑ Continuous physical and DC states during mode transfer.
- ☑ Adaptive switching based on voltage and frequency deviations.

Future Development

- ☐ Establish admissible switching and dwell-time conditions for the hybrid DAE.
- ☐ Simulation on larger-scale power systems.
- ☐ Development of Intelligent switching control to manage transients and reduce switching errors

References

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2. H. Ding, R. Kar, Z. Miao, and L. Fan, "A novel design for switchable grid-following and grid-forming control," *IEEE Trans. Sustainable Energy*, 16(2), 1301–1314, 2025.
3. MATPOWER, *case14: IEEE 14-bus test case*, converted from the IEEE Common Data Format.

Q & A

Thank you for your attention

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